Statistical Modelling and Inference for Health – Assessment 2

1 Introduction

Reducing mortality in critically ill patients remains a primary objective in clinical research. Milnepan, a therapeutic agent of interest, has been hypothesized to significantly decrease the risk of short-term mortality. However, evaluating its true efficacy outside of randomized controlled trials (RCTs) presents substantial challenges due to the inherent biases found in observational data.

The primary objective of this report is to evaluate whether Milnepan is effective in reducing the 7-day risk of death.

2 Data and Methods

2.1 Data Description

We checked the data structure and integrity by *summary*() and *is.na*()

```
summary(heff[,2:12])
missing_pct <- colMeans(is.na(heff[,2:12])) * 100
missing_pct
```

```
    heffpox            sex              age             bmi
 Min.   :0.0000   Min.   :0.0000   Min.   :18.00   Min.   :14.01
 1st Qu.:1.0000   1st Qu.:0.0000   1st Qu.:30.00   1st Qu.:25.27
 Median :1.0000   Median :0.0000   Median :43.00   Median :27.97
 Mean   :0.9216   Mean   :0.4846   Mean   :44.29   Mean   :27.98
 3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:55.00   3rd Qu.:30.69
 Max.   :1.0000   Max.   :1.0000   Max.   :97.00   Max.   :42.98
                                                   NA's   :2914
    smoking          diabetes         milnepan          icu
 Min.   :0.0000   Min.   :0.0000   Min.   :0.000   Min.   :0.00000
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:0.00000
 Median :0.0000   Median :0.0000   Median :1.000   Median :0.00000
 Mean   :0.2479   Mean   :0.1048   Mean   :0.516   Mean   :0.09754
 3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.000   3rd Qu.:0.00000
 Max.   :1.0000   Max.   :1.0000   Max.   :1.000   Max.   :1.00000
 NA's   :406
     TEVENT              Status          death7day
 Min.   : 0.000186   Min.   :0.0000   Min.   :0.0000
 1st Qu.: 3.560166   1st Qu.:0.0000   1st Qu.:0.0000
 Median :10.000000   Median :0.0000   Median :0.0000
 Mean   : 7.119276   Mean   :0.4598   Mean   :0.3719
 3rd Qu.:10.000000   3rd Qu.:1.0000   3rd Qu.:1.0000
 Max.   :10.000000   Max.   :1.0000   Max.   :1.0000

   heffpox        sex        age        bmi    smoking   diabetes   milnepan
 0.000000   0.000000   0.000000  32.782090   4.567443   0.000000   0.000000
       icu     TEVENT     Status  death7day
 0.000000   0.000000   0.000000   0.000000
```

The missing values are mainly concentrated in two variables: bmi and smoking,

BMI(32.78% missing): This is the most serious omission. BMI is often an important risk factor in clinical prediction models (CPM). Such a high proportion of deletions would have resulted in a substantial loss

of samples and potentially significant selection bias if they had been removed directly (Complete Case Analysis).

Smoking(4.57% missing): The missing rate was low. Although the proportion is small, it is preferable to incorporate it into the MI process in order to maintain sample integrity when making causal inferences.

For these 2 deletions, their missing pattern was examined by md.pattern() and md.pairs().

```
pairs_stats <- md.pairs(heff[,2:12])
pairs_stats$rr
round(100 * pairs_stats$mr / (pairs_stats$mr + pairs_stats$mm))
missing_pattern <- md.pattern(heff[,2:12], rotate.names = TRUE)
```

	heffpox	sex	age	bmi	smoking	diabetes	milnepan	icu	TEVENT	Status	death7day
bmi	100	100	100	0	94	100	100	100	100	100	100
smoking	100	100	100	58	0	100	100	100	100	100	100

All the ratios are above 50%, which means that other factors can be used to do the multiple imputation for bmi and smoking.

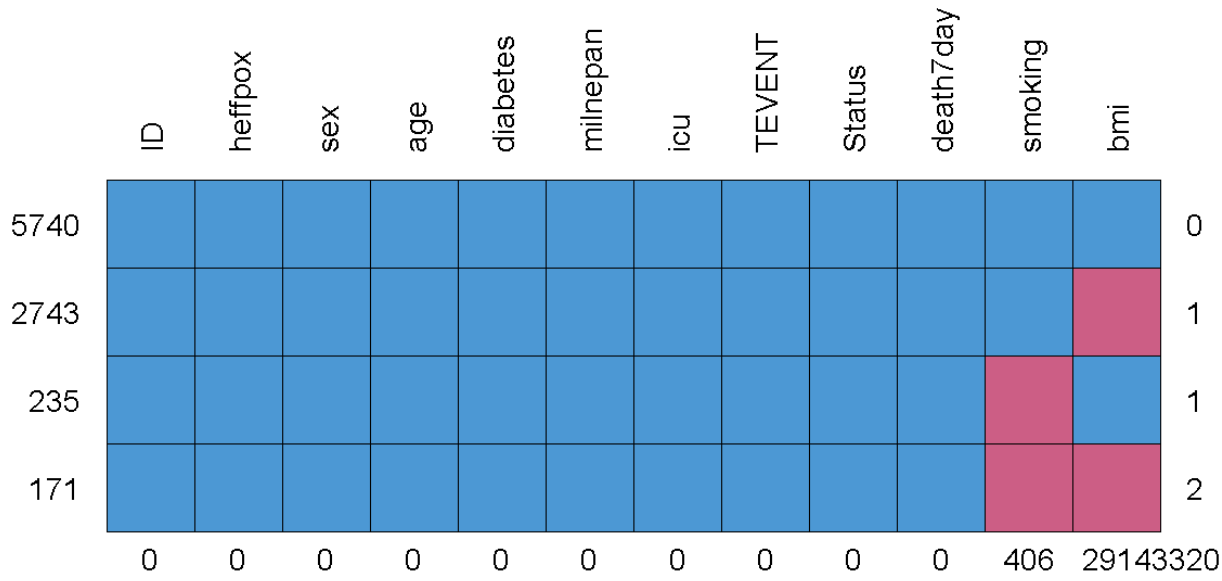

Figure 1. Missing pattern result.

A total of 5,740 samples (approximately 64.6%) were completely complete with no omissions. Of these, 2743 samples were missing only bmi. 235 samples were only missing smoking and 171 samples were missing both. The missing rate of bmi was about 32.8%. Although a little high, it is within the acceptable range of multiple imputation. The absence rate of smoking is very low (about 4.6%), and it can almost be perfectly repaired through multiple interpolations.

2.2 Multiple Imputation

Multiple imputation for bmi and smoking was performed with the mice() function. Before imputation, the column "ID" in the original data set was removed to reduce the influence of irrelevant variables.

```
imputed_data <- mice(heffpox_prep, m = 5, printFlag = FALSE)
densityplot(imputed_data, ~bmi)
stripplot(imputed_data, bmi ~ .imp, pch = 20, cex = 1.2)
```

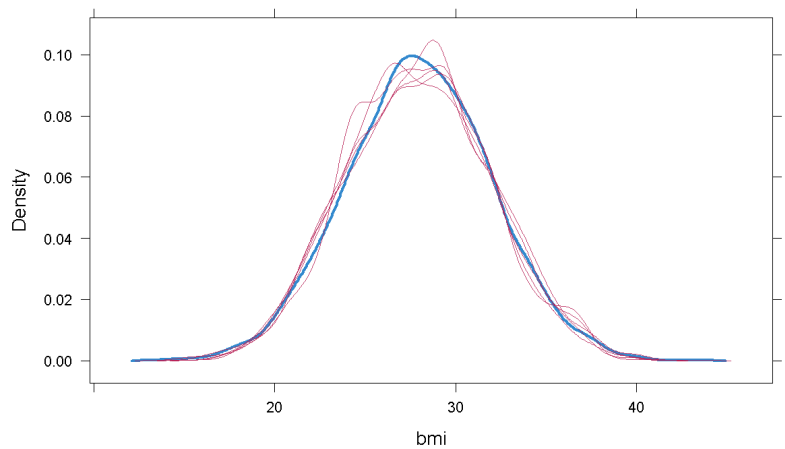

Figure 2. Density plot for bmi: Similar imputed values to the original observations.

The red lines coincide highly with the blue lines, which indicates that the imputed model preserves the distribution characteristics of the original data very well. The imputed BMI values were consistent with the true observed values in mean, variance, and overall morphology without significant shifts.

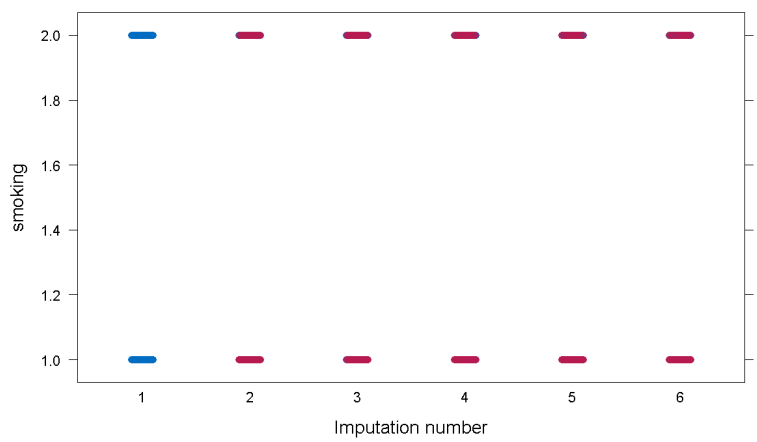

Figure 3. Scatter plot for Smoking: Discreteness of imputation was verified.

All interpolation values (red dots) strictly fall within the horizontal lines of 1.0 and 2.0. This confirms the discreteness of the interpolation value.

After completing the multiple imputation, the full dataset was analyzed by generalized linear models, and the result is generated by Robin's Rule.

```
Coef_values <- vector()
for (i in 1:5) {
  temp_data <- complete(imputed_data, action = i)
  fit <- with(temp_data, glm(death7day ~ milnepan, data = temp_data,
                             family = binomial))
  R <- coef(fit)
  R <- R[2]
  Coef_values[i] <- R}
```

```
[1] -0.5760956 -0.5760956 -0.5760956 -0.5760956 -0.5760956
```

```
fit <- with(imputed_data, glm(death7day ~ milnepan, data = temp_data,
                             family = binomial))
summary(pool(fit))
```

```
         term   estimate  std.error  statistic       df      p.value
1 (Intercept) -0.2372762 0.03070747  -7.726987 8885.001 1.220910e-14
2   milnepan1 -0.5760956 0.04435464 -12.988395 8885.001 3.169214e-38
```

Both result of analysis and pooled results suggested that the interpolation result is very stable. They illustrated that milnepan reduces the risk of death by an average of about 42.4% (p<0.001).

3 Results

3.1 Exploratory Analysis

First, the overall trend of the clinical background of the patients was studied in the first imputed dataset.

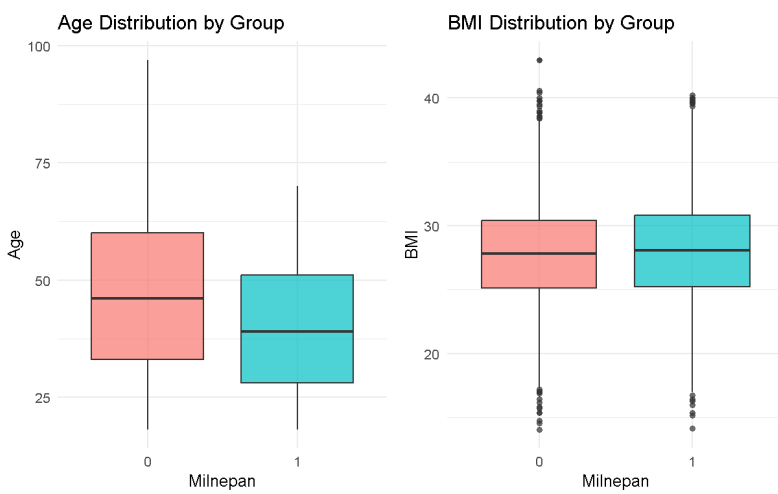

Figure 4. Distribution of age and bmi, grouping by usage of milnepan.

For age, people in the control group are generally older. In medical research, advanced age is often accompanied by a higher underlying risk of death, which can "smear" the performance of the control group. Another problem is that none of the individuals over 70 have taken milnepan.

The median of bmi of both groups was around 28, and the distribution intervals were highly overlapping. Meanwhile, more low BMI outliers emerged in the control group, while both groups had numerous outliers at the high BMI (obesity) end.

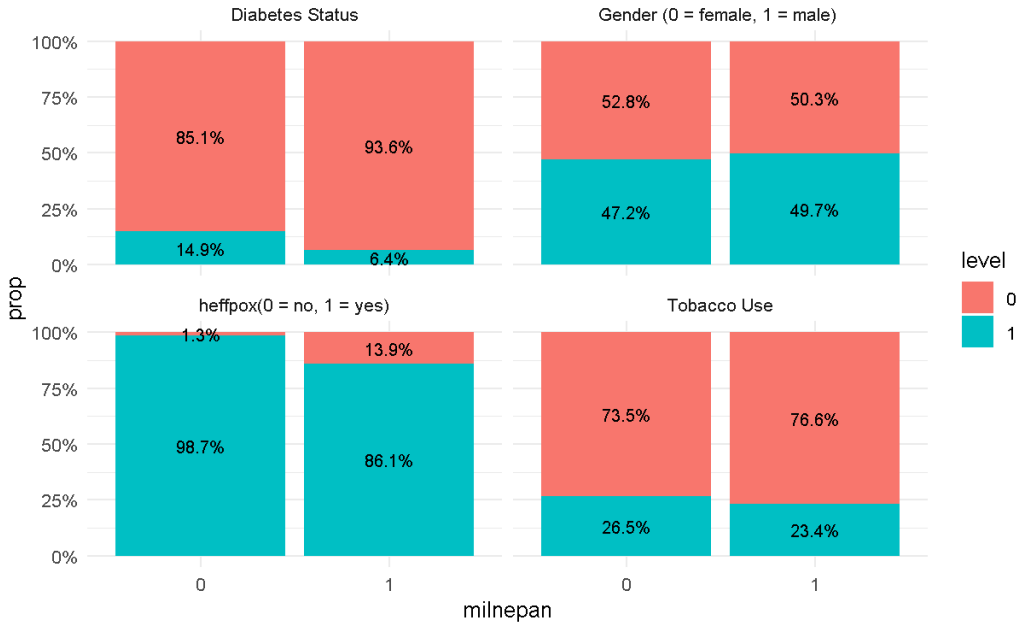

Figure 5. Distribution of diabetes, sex, smoking, and diagnosis of heffpox, grouping by usage of milnepan.

For diabetes status, The prevalence rate of the control group (14.9%) was more than twice that of the milnepan group (6.4%). The metabolic health status of the control group was worse, increasing the confounding risk of death.

13.9% of cases of heffpox had been prescribed milnepan and 1,3% patients without a diagnosis of heffpox also had been prescribed milnepan.

Both control and milnepan groups have similar distributions of sex and smoking factors.

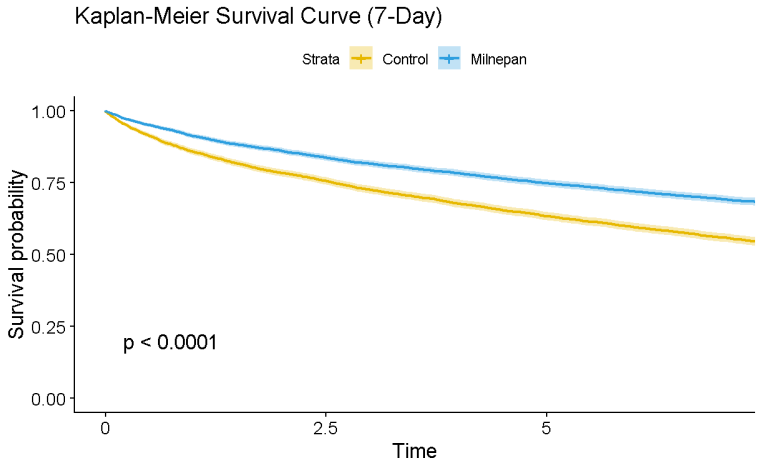

Figure 6. Survival analysis results.

The blue line (Milnepan group) is always above the yellow line (Control group). The medication group showed a significant survival advantage ($p < 0.0001$). This suggests that milnepan is effective in reducing the risk of death before the analysis of confounding factors is carried out.

3.2 Causal Modelling

From previous exploratory analyses, a DAG is drew to show the causal logic among variables.

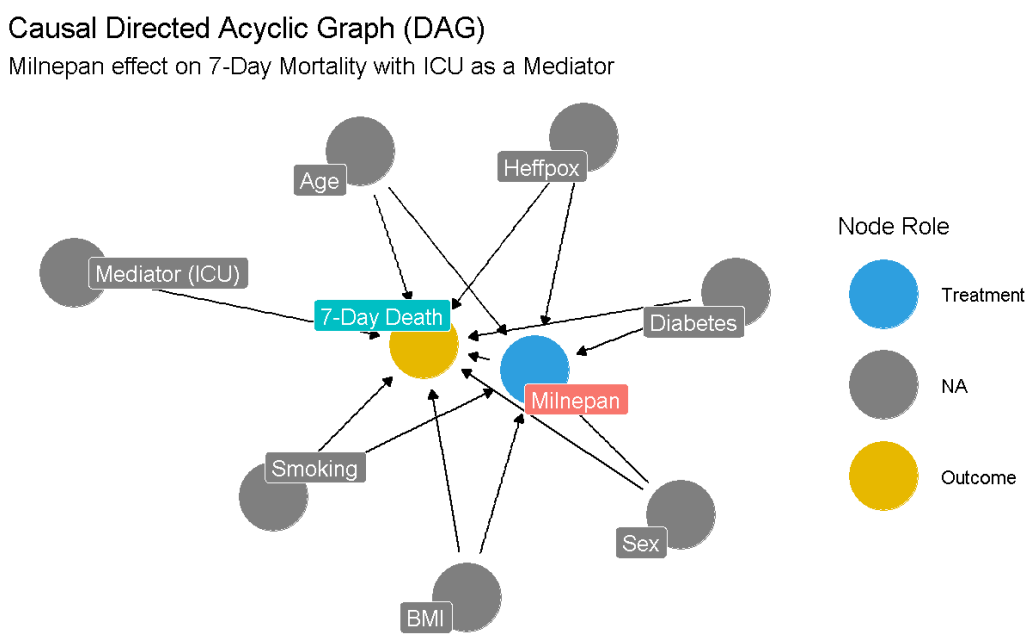

Figure 7. Directed Acyclic Graph (DAG) of all variables.

All variables effect both the usage of milnepan and 7-day death except "icu". It shows that confounders are age, sex, bmi, smoking, diabetes, and heffpox.

Here the hypotheses are formulated as follow:

H_0: Milnepan is ineffective. That is, there was no difference in the risk of death between the medication group and the control group.

H_a: Milnepan is effective. That is, there is a significant difference in the risk of death between the two groups.

3.2.1 Multivariable Regression

```
fit_multivariable <- with(imputed_data, glm(death7day ~ milnepan + age + sex
+ bmi + smoking + diabetes + heffpox, family = binomial))
pooled_multivariable <- pool(fit_multivariable)
summary(pooled_multivariable, exponentiate = TRUE, conf.int = TRUE)
```

```
         term     estimate    std.error  statistic         df
1 (Intercept) 1.365472e-09 1.424958e+02 -0.1432447 8878.98400
2   milnepan1 8.672489e-01 4.880921e-02 -2.9180827 6136.78745
      p.value         2.5 %        97.5 %     conf.low     conf.high
1 8.861002e-01 6.700386e-131 2.782697e+112 6.700386e-131 2.782697e+112
```

```
2 3.534697e-03  7.881141e-01  9.543296e-01  7.881141e-01  9.543296e-01
```

After accounting for multiple imputation and adjusting for baseline confounders using a pooled multivariable logistic regression model, Milnepan demonstrated a significant protective effect. The adjusted odds ratio (aOR) for 7-day mortality was 0.867 (95% CI: 0.788–0.954, $P = 0.004$). This rejects H_0 and confirms that the observed benefit of Milnepan is not merely due to the treatment group being younger or healthier, but represents a genuine reduction in mortality risk.

Compared with the regression results before, the risk for death was corrected from a 44% reduction to a 13% reduction with milnepan.The previous "super strong drug effect" was largely due to the fact that the patients in the control group were in poorer health. After correction, the efficacy of the drug returned to a more genuine and reasonable level.

3.2.2 Inverse Probability of Treatment Weighting

Firstly, the propensity score model was constructed. According to the previous EDA results bmi was relatively evenly distributed in the two groups, so model simplification and effect comparison were carried out here.

```
fit_full <- with(imputed_data, glm(milnepan ~ age + sex + bmi + smoking
                + diabetes + heffpox, family = binomial))
fit_reduced <- with(imputed_data, glm(milnepan ~ age + sex + smoking
                + diabetes + heffpox, family = binomial))
anova(fit_reduced$analyses[[1]], fit_full$analyses[[1]], test = "LRT")
```

```
Analysis of Deviance Table

Model 1: milnepan ~ age + sex + smoking + diabetes + heffpox
Model 2: milnepan ~ age + sex + bmi + smoking + diabetes + heffpox
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      8883      10944
2      8882      10944  1  0.89022   0.3454
```

I performed a Likelihood Ratio Test (LRT) to evaluate whether bmi should be included in the propensity score model. While the result was not statistically significant ($P = 0.345$), I opted for the Full Model (Model 2) for the final analysis, since prioritizes adjusting for all suspected confounders based on prior knowledge, as the cost of omitting a potential confounder outweighs the cost of adjusting for a non-significant one.

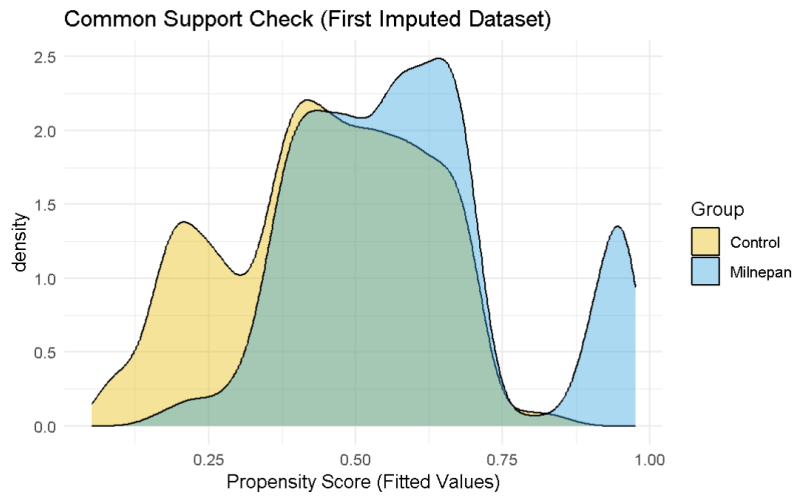

Figure 8. The overlap of propensity scores of the first imputed dataset.

In the control group, a large proportion of patients had a predicted probability of receiving treatment below 0.3, and this proportion of patients was almost nonexistent. In contrast, there was one treated patient with a cluster propensity score higher than 0.9, but no comparable control. This lack of overlap mainly caused by age differences.

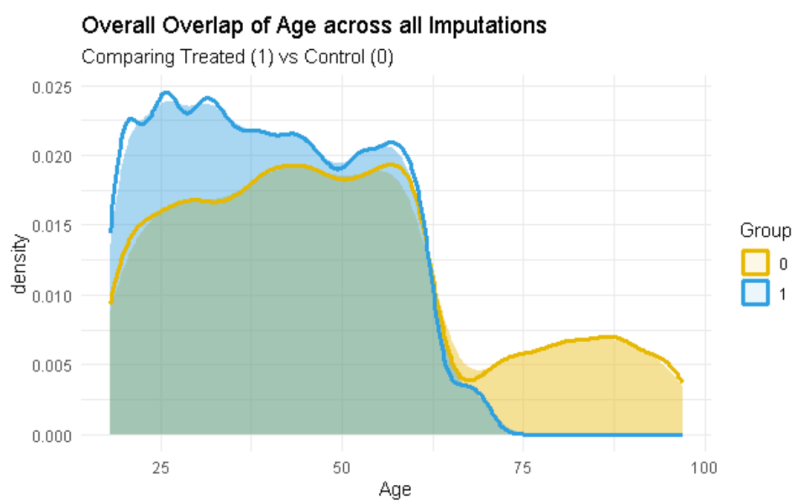

Figure 9. The density plot of age. No one over 75 prescribe milnepan.

Due to limited common support in propensity scores, traditional 1:1 matching failed to achieve covariate balance, particularly for age. I therefore transitioned to Inverse Probability of Treatment Weighting.

```
weighted_res <- weightthem(
  milnepan ~ age + sex + bmi + smoking + diabetes + heffpox,
  datasets = imputed_data, approach = "within",
  method = "ps", estimand = "ATE")
love.plot(weighted_res, thresholds = c(m = 0.1), abs = TRUE)
summary(weighted_res)
```

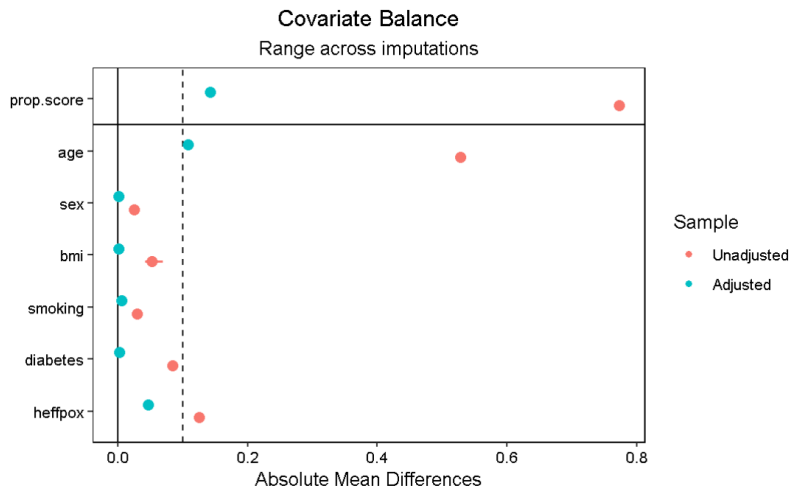

Figure 10. Result of IPTW.

```
- Effective Sample Sizes:

           Control  Treated
Unweighted 4302.    4587.
Weighted   3822.59  4080.61
```

The blue dot (Adjusted) of age has been significantly pulled to the left and is currently hovering around the 0.1 dotted line. Although it has not yet completely dropped below 0.1, it is already on the statistically acceptable edge.

The blue dots for sex, bmi, smoking and diabetes almost all fall on the 0 mark. This means that the weighting method successfully made the treatment group and the control group "completely consistent" in these characteristics.

At the same time, the sample size was only lost by approximately 11%, and nearly 8,000 effective sample sizes were retained.

Given a convincing result of IPTW, I ran the regression model on the weighted data.

```
fit_full_w <- with(weighted_res,
                   svyglm(death7day ~ milnepan + age + sex + bmi
                          + smoking + diabetes + heffpox,
                          family = quasibinomial))
fit_pooled_w <- pool(fit_full_w)
summary(fit_pooled_w, exponentiate = TRUE, conf.int = TRUE)
```

```
         term     estimate   std.error    statistic       df      p.value
1 (Intercept) 1.026350e-09 0.226964216 -91.19172117 6610.020 0.000000e+00
2   milnepan1 8.968200e-01 0.048882355  -2.22780004 7137.939 2.592490e-02
3         age 1.023953e+00 0.001492671  15.85796798 7190.115 1.080303e-55
4        sex1 1.003316e+00 0.048966245   0.06760421 6501.756 9.461028e-01
5         bmi 1.067474e+00 0.006124514  10.66127049 3907.354 3.525533e-26
6    smoking1 1.070366e+00 0.057456136   1.18352477 5987.108 2.366482e-01
7   diabetes1 2.511661e+00 0.083888059  10.97825209 8868.187 7.345362e-28
8     heffpox 3.588999e+07 0.090488273 192.24556486 2096.117 0.000000e+00
```

```
          2.5 %       97.5 %     conf.low     conf.high
1 6.577602e-10 1.601485e-09 6.577602e-10 1.601485e-09
2 8.148722e-01 9.870089e-01 8.148722e-01 9.870089e-01
3 1.020961e+00 1.026954e+00 1.020961e+00 1.026954e+00
4 9.114855e-01 1.104398e+00 9.114855e-01 1.104398e+00
5 1.054733e+00 1.080369e+00 1.054733e+00 1.080369e+00
6 9.563475e-01 1.197978e+00 9.563475e-01 1.197978e+00
7 2.130814e+00 2.960577e+00 2.130814e+00 2.960577e+00
8 3.005420e+07 4.285896e+07 3.005420e+07 4.285896e+07
```

The pooled analysis of 5 imputed datasets showed that milnepan was associated with an Odds Ratio (OR) of 0.897 for 7-day mortality (95% CI:0.815-0.987). This also rejects H_0 and indicates that milnepan is effective in reducing the risk of death in this patient population. This is corresponded to the result of multivariable regression.

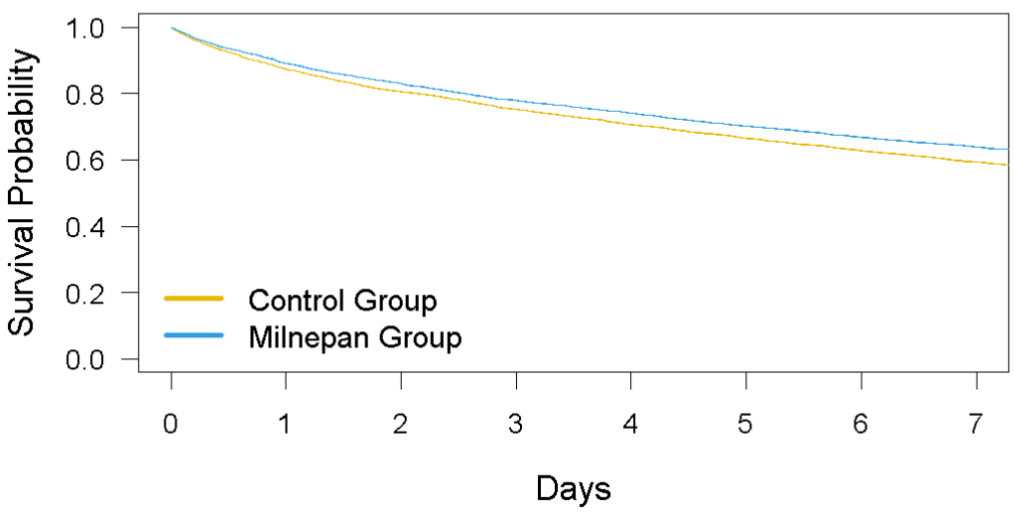

Figure 11. 7-Day Survival: Milnepan vs Control (IPTW Adjusted).

As shown in the survival curves, the probability of survival was consistently higher in the Milnepan group than in the control group. This was confirmed by the Cox model adjusted for IPTW, with a HR of 0.86 (95% CI: 0.80 to 0.92; $P < 0.0001$), indicating a significant risk reduction by the drug.

4 Conclusion

Both multivariable regression and IPTW conducted in the multiple imputed datasets reject the H_0 that milnepan is ineffective. This is also supported by the survival analysis. It can be concluded that milnepan is effective in reducing the risk of death.